

Front-end Essentials

Bootstrap 4



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Lesson Objectives

- Understand basic concepts of Bootstrap 4
- Able to create layout using Bootstrap 4
- Able to use Components of Bootstrap 4 to create complex web app

Section 1

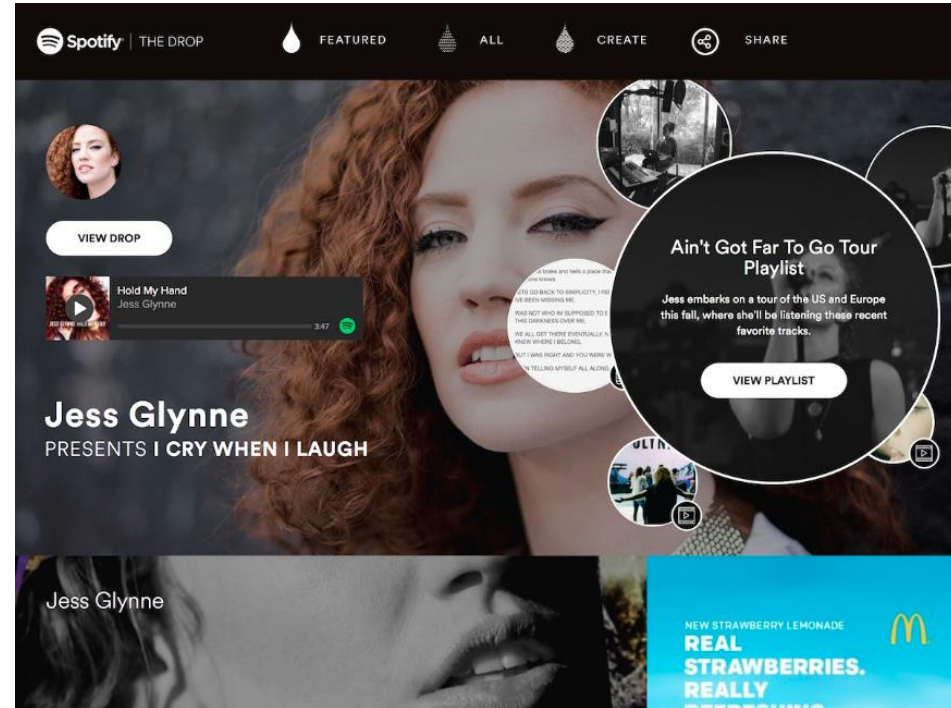
Overview

Overview – What is Bootstrap?

- CSS framework for **faster** and **easier** web development.
- Includes HTML and CSS based design templates.
- Easily create responsive designs.

Overview – Why Bootstrap?

- Beautiful, Modern Design
- Browser support
- Easy to learn
- Responsive Web Design
- Design faster



Overview – How to use Bootstrap?

- **Step 1:** Download Bootstrap at <https://getbootstrap.com/>
- **Step 2:** Add the HTML5 doctype
- **Step 3:** Add meta tag and links

- CSS framework for **faster** and **easier** web development.
- **Benefits:** Beautiful and Modern Design, Browser support, Easy to learn, Responsive Web Design, Design faster
- Start using Bootstrap 4 by download and add it to html page

Section 2

Layout

- Important in Web design
- Help to create a skeleton for our web page



- Syntax:

```
<div class="container">  
  <!-- Content here -->  
</div>
```

```
<div class="container-fluid">  
  ...  
</div>
```

- Use `.container-fluid` for a full width container, spanning the entire width of the viewport

- **Problem:** How to create vertically align block ?
- **Simple Solution:** Use CSS `float` or `position` property
- **With Bootstrap:** Use Grid System

```
<div class="container">  
  <div class="row">  
    <div class="col-sm">  
      One of three columns  
    </div>  
    <div class="col-sm">  
      One of three columns  
    </div>  
    <div class="col-sm">  
      One of three columns  
    </div>  
  </div>  
</div>
```

One of three columns

One of three columns

One of three columns

■ Syntax: `.col-a-b`

- ✓ a: symbol of device (`sm`, `md`, `lg`, `xl`)
- ✓ b: number of columns (`1`, `2`, `3` up to `12`)

	Extra small <576px	Small ≥576px	Medium ≥768px	Large ≥992px	Extra large ≥1200px
Max container width	None (auto)	540px	720px	960px	1140px
Class prefix	<code>.col-</code>	<code>.col-sm-</code>	<code>.col-md-</code>	<code>.col-lg-</code>	<code>.col-xl-</code>
# of columns	12				
Gutter width	30px (15px on each side of a column)				
Nestable	Yes				
Column ordering	Yes				

- Use `.container` and `.container-fluid` to create frame/skeleton for you web page
- Create complex layout with **Grid System**
- **Grid System** consists of **row** where each row is composed of 12 **columns**
- Use class prefix `.col`, `.col-sm-`, `.col-md-`, `.col-lg-`, `.col-xl-` to create responsive web app

Section 3

Typography

- All HTML headings, `<h1>` through `<h6>`, are available

Heading	Example
<code><h1></h1></code>	h1. Bootstrap heading
<code><h2></h2></code>	h2. Bootstrap heading
<code><h3></h3></code>	h3. Bootstrap heading
<code><h4></h4></code>	h4. Bootstrap heading
<code><h5></h5></code>	h5. Bootstrap heading
<code><h6></h6></code>	h6. Bootstrap heading

- Classes **.h1** through **.h6** are also available

h1. Bootstrap heading

h2. Bootstrap heading

h3. Bootstrap heading

h4. Bootstrap heading

h5. Bootstrap heading

h6. Bootstrap heading

```
<p class="h1">h1. Bootstrap heading</p>
<p class="h2">h2. Bootstrap heading</p>
<p class="h3">h3. Bootstrap heading</p>
<p class="h4">h4. Bootstrap heading</p>
<p class="h5">h5. Bootstrap heading</p>
<p class="h6">h6. Bootstrap heading</p>
```

Copy

- Easily realign text to components with text alignment classes

Left aligned text on all viewport sizes.

Center aligned text on all viewport sizes.

Right aligned text on all viewport sizes.

Left aligned text on viewports sized SM (small) or wider.

Left aligned text on viewports sized MD (medium) or wider.

Left aligned text on viewports sized LG (large) or wider.

Left aligned text on viewports sized XL (extra-large) or wider.

```
<p class="text-left">Left aligned text on all viewport sizes.</p>
<p class="text-center">Center aligned text on all viewport sizes.</p>
<p class="text-right">Right aligned text on all viewport sizes.</p>

<p class="text-sm-left">Left aligned text on viewports sized SM (small) or wider.
<p class="text-md-left">Left aligned text on viewports sized MD (medium) or wider
<p class="text-lg-left">Left aligned text on viewports sized LG (large) or wider.
<p class="text-xl-left">Left aligned text on viewports sized XL (extra-large) or
```

Copy

- Transform text in components with text capitalization classes

lowercased text.

UPPERCASED TEXT.

CapiTaliZed Text.

```
<p class="text-lowercase">Lowercased text.</p>  
<p class="text-uppercase">Uppercased text.</p>  
<p class="text-capitalize">CapiTaliZed text.</p>
```

Copy

- **Unstyled:** Remove the default **list-style** and left margin on list items

```
<ul class="list-unstyled">
  <li>Lorem ipsum dolor sit amet</li>
  <li>Consectetur adipiscing elit</li>
  <li>Integer molestie lorem at massa</li>
  <li>Facilisis in pretium nisl aliquet</li>
  <li>Nulla volutpat aliquam velit
    <ul>
      <li>Phasellus iaculis neque</li>
      <li>Purus sodales ultricies</li>
      <li>Vestibulum laoreet porttitor sem</li>
      <li>Ac tristique libero volutpat at</li>
    </ul>
  </li>
  <li>Faucibus porta lacus fringilla vel</li>
  <li>Aenean sit amet erat nunc</li>
  <li>Eget porttitor lorem</li>
</ul>
```

Lorem ipsum dolor sit amet
Consectetur adipiscing elit
Integer molestie lorem at massa
Facilisis in pretium nisl aliquet
Nulla volutpat aliquam velit

- Phasellus iaculis neque
- Purus sodales ultricies
- Vestibulum laoreet porttitor sem
- Ac tristique libero volutpat at

Faucibus porta lacus fringilla vel
Aenean sit amet erat nunc
Eget porttitor lorem

- **Inline:** Remove a list's bullets and apply some light **margin** with a combination of two classes, **.list-inline** and **.list-inline-item**

Lorem ipsum Phasellus iaculis Nulla volutpat

```
<ul class="list-inline">  
  <li class="list-inline-item">Lorem ipsum</li>  
  <li class="list-inline-item">Phasellus iaculis</li>  
  <li class="list-inline-item">Nulla volutpat</li>  
</ul>
```

- Bootstrap sets basic global display, typography, and link styles
- Change text alignment, transform, style, weight, and color with our [text utilities](#) and [color utilities](#).

Section 4

Common Components

- Navigation Bar
- Breadcrumb
- Table
- Forms
- Buttons
- Images
- Icons
- Dropdown
- Pagination
- Label
- Badges
- Alert
- Progress bar
- List Group

- **Navigation Bar** is a section of a graphical user interface intended to aid visitors in accessing information
- Require a wrapping `.navbar` with `.navbar-expand{-sm|-md|-lg|-xl}`

Navbar Home Link Dropdown ▼ Disabled

- Indicate the current page's location within a navigational hierarchy that automatically adds separators via CSS.

Home

[Home](#) / Library

[Home](#) / [Library](#) / Data

- **Table** represents tabular data
- **Bootstrap 4** support for table by adding `.table` class to any `<table>` element

#	First	Last	Handle
1	Mark	Otto	@mdo
2	Jacob	Thornton	@fat
3	Larry the Bird		@twitter

- **Bootstrap** provides several form control styles, layout options, and custom components for creating a wide variety of forms.

Email address

We'll never share your email with anyone else.

Password

☐ Check me out

Submit

- **Form controls:** Textual form controls—like `<input>`s, `<select>`s, and `<textarea>`s—are styled with the `.form-control`
- Default checkboxes and radios are improved upon with the help of `.form-check`

- **Bootstrap** includes several predefined button styles, each serving its own semantic purpose
- Add `.btn-lg` or `.btn-sm` for additional sizes
- Make buttons look inactive by adding the disabled `boolean` attribute to any `<button>` element



- Images in Bootstrap are made responsive with `.img-fluid` (`max-width: 100%;` and `height: auto;`)



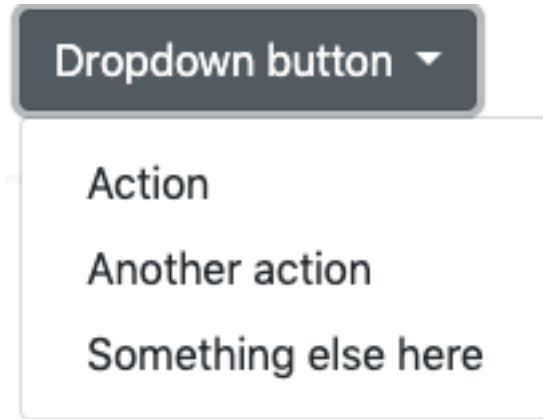
```

```

Copy

- **Bootstrap** doesn't include an icon library by default
- Preferred:
 - ✓ Iconic
 - ✓ Octicons

- Dropdowns are toggleable, contextual overlays for displaying lists of links and more
- JavaScript is **required** for Dropdown



- Wrap the dropdown's toggle (your button or link) and the dropdown menu within **.dropdown**
- Add **.dropdown-item** to each item and **.dropdown-menu** to the menu

```
<div class="dropdown">
  <button class="btn btn-secondary dropdown-toggle" type="button" id="dropdownMenu1">
    Dropdown button
  </button>
  <div class="dropdown-menu" aria-labelledby="dropdownMenuButton">
    <a class="dropdown-item" href="#">Action</a>
    <a class="dropdown-item" href="#">Another action</a>
    <a class="dropdown-item" href="#">Something else here</a>
  </div>
</div>
```

Copy

- **Bootstrap** provides many UI Components that help you to developer faster and easier
- To use a component you must specify correct Bootstrap syntax with corresponding class
- Most Components only need HTML and CSS
- For Dropdown you must add JavaScript

Thank you

